

AirSense Pakistan

Vendor Pricing Report

Hardware Procurement - Campus Pilot Phase 1
Beaconhouse International College, Islamabad

Prepared By	AirSense Pakistan Technical Team
Date	May 2026
Total Budget Requested	PKR 13,163 (One-Time Hardware Procurement)
Procurement Strategy	Local Pakistan market
Procurement Timeline	3 to 7 business days for complete delivery (all vendors local or domestic e-commerce)

1. Vendor Overview

All hardware components in this budget are sourced from Pakistan-local vendors. This eliminates international shipping delays, customs duties, and import uncertainty. The two primary procurement channels are Daraz.pk (Pakistan's largest e-commerce marketplace) and Hallroad Electronics Market, Lahore (a well-established electronics wholesale district). Supplementary items are sourced from local hardware stores in Islamabad.

Vendor	Type	Est. Delivery	Reliability	Best Used For
Daraz.pk/Lahorestore.pk	National E-commerce	2 to 5 business days	High	PM sensor, microSD, jumper wires, enclosure accessories
Hallroad Lahore	Electronics Wholesale	Same day / 1 to 2 days	High	ESP32, BME280, SD module, rain sensor, power supply
Epro Pakistan	Online Electronics	3 to 5 business days	Medium-High	ESP32 boards (backup source if Hallroad delayed)
Local Hardware Store	Hardware / Electrical	Same day	High	Extension cable, silicone sealant, mounting fasteners
International (AliExpress)	International Import	10 to 25 business days	Not Recommended	Backup only if local stock unavailable

2. Complete Line-Item Pricing

2.1 Core Build (Original 12 Components)

These twelve items comprise the original core build as specified in the revised PM2.5 Forecasting Budget and Framework document. All prices are sourced from Pakistan local markets as of 2025/2026.

#	Component / Item	Model / Specification	Vendor	Qty	Unit (PKR)	Total (PKR)
1	PM2.5/PM10/PM1 Sensor	PMS7003 Laser Particle Sensor	Daraz.pk/Lahorestore.pk	1	4,589	4,589
2	Microcontroller	ESP32 WROOM-32D Dev Board	Hallroad Lahore / Epro	1	1,150	1,150
3	Environmental Sensor	BME280 Temp/Humidity/Pressure	Hallroad Lahore	1	650	650
4	Data Logging Module	Arduino MicroSD Card Reader	Hallroad Lahore	1	147	147
5	Storage Media	Samsung EVO Plus 16 GB MicroSD	Daraz.pk	1	989	989
6	Weather Enclosure	Waterproof DIY Enclosure (IP65)	Hallroad Lahore	1	343	343
7	Jumper Wires	20 cm Mixed 40-piece Kit	Daraz.pk	1	199	199
8	Power Supply	5V 2A with Cable	Hallroad Lahore	1	294	294
9	Cable Ties	100 pcs Assorted Pack	Daraz.pk	1	100	100
10	Hose Clamps	Pack of 2 Stainless 2-inch	Daraz.pk	1	139	139
11	Radiation Shield	White Plastic Plate Set	Daraz.pk	1	299	299
12	Rain Proxy Sensor	Raindrop Detection Module	Hallroad Lahore	1	147	147
CORE BUILD SUBTOTAL					PKR 9,046	

2.2 Recommended Additions

These four items are additions recommended by the AirSense technical team to ensure a robust, complete campus rooftop deployment. Items 13 and 14 were also part of the original framework document (anemometer). Items 15 and 16 are new additions identified during hardware sufficiency review.

#	Component / Item	Model / Specification	Vendor	Qty	Unit (PKR)	Total (PKR)
13	Validation Anemometer	Handheld Anemometer (Field Use)	Hallroad / Online	1	2,090	2,090
14	Power Extension Cable	10 m Heavy-Duty 3-Pin Extension	Local Hardware Store	1	800	800
15	Weatherproof Sealant	Neutral-Cure Silicone (300 ml)	Local Hardware Store	1	350	350
16	Mounting Fasteners	M4/M6 Stainless Bolt Set	Local Hardware Store	1	250	250

ADDITIONS SUBTOTAL	PKR 3,490
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HARDWARE SUBTOTAL (Items 1 to 16)	PKR 12,536
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#	Component / Item	Model / Specification	Vendor	Qty	Unit (PKR)	Total (PKR)
17	Contingency Reserve	5% of hardware total for spares and consumables	—	1	627	627

TOTAL BUDGET REQUEST	PKR 13,163
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3. Procurement Notes and Sourcing Guidance

3.1 PMS7003 PM Sensor (PKR 4,589)

The PMS7003 is the most critical and most expensive single component. Confirm the listing specifically states PMS7003 (not G5, G7, or other PMS series variants). The PMS7003 outputs PM1, PM2.5, and PM10 simultaneously over UART, which is required for the feature set. Order from Daraz.pk/Lahorestore.pk as it is reliably stocked; confirm seller rating above 95% positive before ordering.

3.2 ESP32 WROOM-32D (PKR 1,150)

The WROOM-32D variant is preferred as it is the most widely stocked and has the best community library support. WROOM-32 (without the D suffix) is also acceptable if the D variant is not in stock. Do not substitute with ESP8266 — it lacks sufficient GPIO pins and processing capacity for this application. Hallroad Lahore will have this in stock same-day; Epro.pk is a reliable online backup.

3.3 BME280 (PKR 650)

Confirm the listing specifies BME280, not BMP280. The BMP280 does not measure humidity and is a common look-alike product. Both measure temperature and pressure, but humidity is a required feature in the forecasting framework. Hallroad Lahore is the most reliable local source. If ordering online, verify the module has the BME280 marking on the chip.

3.4 Weatherproof Enclosure (PKR 343)

The enclosure must be genuinely waterproof — an IP65 or higher rating is required for outdoor rooftop use. Many generic plastic boxes marketed as 'waterproof' do not hold their seal in rain. Verify the listing includes a rubber gasket around the lid and gland fittings or at minimum accepts silicone-sealed cable pass-throughs. The silicone sealant addition (item 15) is specifically included to supplement the enclosure's cable entry sealing.

3.5 Handheld Anemometer (PKR 2,090)

The anemometer does not need to be a precision meteorological instrument. A standard digital handheld unit measuring wind speed in metres per second or kilometres per hour is adequate for field validation purposes. The team will use it during monthly maintenance visits to compare spot-measured wind speed with the public API wind speed for the same location and time. A difference of less than 2 m/s is acceptable; larger persistent differences would indicate the API data source needs to be recalibrated or replaced.

3.6 Extension Cable (PKR 800)

A rooftop installation exposes the cable to UV radiation, heat, and rain. A standard domestic extension cable is not rated for permanent outdoor use. Specify a heavy-duty outdoor extension cable with UV-resistant outer insulation, or run the cable through a conduit (cable tray or flexible metallic conduit) from the indoor socket to the unit. A 10-metre length covers most rooftop scenarios; verify the rooftop socket distance during the pre-deployment site survey.

4. Procurement Timeline

Day	Activity	Vendor	Notes
Day 1	Submit ecommerce orders (items 1, 5, 7, 9, 10, 11)	Daraz.pk	Order online. Delivery 2 to 5 business days.
Day 1	Order from Hallroad Lahore (items 2, 3, 4, 6, 8, 12)	Hallroad Lahore	Same-day procurement. Test components before leaving market.
Day 2	Purchase local hardware items (items 14, 15, 16)	Local Hardware Store	Same-day. Any hardware or electrical shop in Islamabad.
Day 3 to 5	ecommerce orders arrive. Source anemometer (item 13)	Daraz / Hallroad	Purchase anemometer from Hallroad or online if not yet sourced.
Day 6 to 7	Assemble and bench-test sensor circuit	On-site (lab)	Full integration test before rooftop deployment.
Day 8	Rooftop installation and commissioning	Campus rooftop	Install, connect, verify live data flow to cloud.

5. Cost Context and Value Assessment

For reference, the table below contextualises the proposed hardware cost against commercial alternatives and the value the pilot is intended to generate:

Option	Cost (PKR)	Notes
This Proposal (AirSense Build)	13,163	Complete rooftop forecasting sensor with 16-component validated stack
Commercial IoT PM Monitor (import)	60,000 to 200,000	Typical commercial monitors do not include forecasting or API integration
PurpleAir PA-II (import, USD 250+)	70,000+	US-based sensor; no local support; high customs duties likely
Government EPA Monitor (reference)	500,000+	Regulatory-grade; no forecasting component
AirSense Pilot Value (3-month data)	Training dataset worth 50,000+ in commercial terms	Real-world Islamabad data directly enables improved forecasting accuracy for enterprise product

Summary

At PKR 13,163, this is one of the most cost-efficient AI/IoT research deployments possible. The build uses tested components available in the local market, requires no import duties or international shipping, and produces a dataset that directly underpins the commercial value of the AirSense platform.